

Name: IAN LEONG ZHENG YAN

StudentID: 34423680

Cruise ID: 1234

Cruise ID 1234
Cruise Name Sydney Brisbane Getaway
Cost per person \$810

Port Code	Port Name	Port Latitude	Port Longitude	Country code	Country Name	Departure/Arrival	Date/Time
AUSYD	Sydney	-33.865143	151.209900	AUS	Australia	Depart	19 Feb 2025 15:30
AUBNE	Brisbane	-27.470125	153.021072	AUS	Australia	Arrive	21 Feb 2025 07:00
AUBNE	Brisbane	-27.470125	153.021072	AUS	Australia	Depart	21 Feb 2025 17:00
AUSYD	Sydney	-33.865143	151.209900	AUS	Australia	Arrive	23 Feb 2025 08:00

1NF

CRUISE (cruise_id, cruise_name, cruise_cost (port_code, port_name, port_latitude, port_longitude, country_code, country_name, port_datetime))

1NF

CRUISE (cruise_id, cruise_name, cruise_cost)

CK: cruise_id

PORT_SCHEDULE (cruise_id, port_datetime, port_code, port_name, port_latitude, port_longitude, country_code, country_name)

CKs: cruise_id, port_datetime

No Partial Dependency

2NF

CRUISE (cruise_id, cruise_name, cruise_cost)

CK: cruise_id

PORT_SCHEDULE (cruise_id, port_datetime, port_code, port_name, port_latitude, port_longitude, country_code, country_name)

CKs: cruise_id, port_datetime

Transitive Dependency:

port_code → port_name, port_latitude, port_longitude,
country_code → country_name

3NF

CRUISE (cruise_id, cruise_name, cruise_cost)

PORT_SCHEDULE (cruise_id, port_datetime)

PORT (port_code, port_name, port_latitude, port_longitude, country_code)

COUNTRY (country_code, country_name)

Full Dependencies:

cruise_id → cruise_name, cruise_cost

port_code → port_name, port_latitude, port_longitude, country_code

country_code → country_name